

Jeevan Sanchez

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EDUCATION

Queen's University

Bachelor of Applied Science in Computer Engineering

- Direct Entry Computer Engineering

Exp. May 2029

Kingston, ON

EXPERIENCE

Queen's Rocket Engineering Team

Propulsion Integration Systems Designer

Avionics Lead (since September 2026)

September 2025 - Present

Kingston, ON

- **Co-leading** the avionics subteam while driving **systems design** and **hardware development** across propulsion, sensing, and communication systems.
- **Designed** two **low-EMI 4-layer control and sensing avionics PCBs** integrating **analog signal conditioning**, solenoid actuation, **power management**, and **STM32/ESP32** interfaces.
- **Authored embedded C/C++ firmware drivers** for precision **ADCs**, Hall-effect sensors, and real-time actuator control over **SPI, I²C, and CAN/TWAI** buses.
- **Executing** board bring-up, **hardware debugging**, and troubleshooting through **hand-soldering, rework, bench testing, and functional integration testing**.

Queen's Engineering Society IT-OPS Team

IT-Operations Team Member

September 2025 - May 2026

Kingston, ON

- **Provide first-level technical support** within a small team maintaining Engineering Society production systems, troubleshooting **hardware/software issues** and managing **support tickets**.
- **Researched an ML-powered RAG pipeline** using **LangChain and OpenAI** to automate IT support workflows and improve **knowledge retrieval and ticket resolution**.

PROJECTS

Penguin ([Link](#))

Intelligent Robotics | ESP32-S3, C++, Arduino, PlatformIO, KiCad, OpenCV, Next.js)

May 2026 - Present

Toronto, ON

- **Designed** a low-EMI 4-layer (4L) **ESP32-S3-1U controller PCB** for a **hybrid wheel-legged rover**, utilizing **continuous copper pours, stitching vias, controlled-impedance trace routing**, and raw driver breakout **IC interfaces** for the **OV2640 DVP camera, IMU, and motor drivers**.
- **Engineered** a full power architecture regulation system for the rover, integrating a 3-stage regulation cascade dedicated to camera power delivery, CMOS-controlled power distribution, and **BMS** protection.
- **Authored** modular C++ firmware in **PlatformIO** with custom peripheral drivers for the **IMU** and motor drivers, executing real-time **PID-based locomotion** control loops and **OV2640** frame acquisition.
- **Built** real-time **HTTP/WebSocket** telemetry and control streaming paired with a **Next.js/Three.js** web dashboard, while actively **integrating OpenCV** for real-time **computer vision perception**.

Vireo ([Link](#))

September 2026 - Present

Audio Interface, Signal Processing | (KiCad, STM32, C/C++, Analog/Digital Electronics)

Toronto, ON

- **Designing a multi-layer, low-noise audio interface PCB** for multi-instrument acquisition , integrating custom **analog signal conditioning**, high-resolution **audio ADCs**, split-rail **power management**, and **ESD/EMI-protected I/O** stages.
- **Engineering precision analog signal chains** featuring **active gain stages**, **anti-aliasing filters**, **impedance matching**, and **mixed-signal PCB layout strategies** (isolated analog/digital ground planes and continuous copper pours) to **eliminate cross-talk** and **thermal/electromagnetic noise**.
- **Developing low-latency embedded C/C++ firmware** on **STM32** utilizing peripheral **I2S/SAI** and **DMA** ping-pong buffers for **multi-channel audio acquisition**, real-time **digital signal processing (DSP)**, device control, and hardware **MIDI** communication.

HomeCtrl ([Link](#))

August 2025 - July 2026

Smart Home-Automation Controller | (ESP32, KiCad, WebSocket, Arduino C/C++, Next.js)

Toronto, ON

- **Engineered schematics** and a custom 4-layer (4L) **ESP32-S3** main **controller PCB** and a companion **I²C** light-sensor breakout board featuring a **controlled-impedance stackup**, continuous **copper pours**, solid **ground planes**, localized **decoupling capacitance**, **PDN** optimization, and **ESD/EMI-protected** driver stages for pumps and steppers.
- **Authored modular C/C++ firmware** in **PlatformIO** running on **FreeRTOS**, orchestrating concurrent tasks, **queues**, and **mutexes** for real-time **actuator control**, **analog/digital sensor acquisition**, and deterministic **finite state machine (FSM)** execution.
- **Developed Wi-Fi/WebSocket telemetry and control pipelines** linked to a **Next.js** dashboard and **Alexa** voice control for end-to-end system validation.

CERTIFICATIONS

Microsoft Azure AI Fundamentals

Microsoft | Azure, AI/ML | AI-900

August 2025

Toronto, ON

CS50's Introduction to Artificial Intelligence with Python

Harvard University | AI/ML Theory, Python, sklearn | CS50AI

July 2025

Toronto, ON

CS50's Introduction to Computer Science

Harvard University | C Programming, Python, JavaScript, SQL | CS50

July 2024

Toronto, ON

AWARDS

First Place at UofT UTRA Hacks Hackathon

Health System with Integrated Robotics and Firmware | (Arduino C/C++, Next.js, AI/ML)
University of Toronto

February 2025

Toronto, ON

First Place at UofT SEEK JR. Hackathon

Autonomous Martian Rover | (Arduino C/C++)
University of Toronto

February 2024

Toronto, ON